

# Playwright

XPATH-Axes

How? When?

# The Golden Rule

What

What is Xpath-Axes?  
XML path based on relationship

When

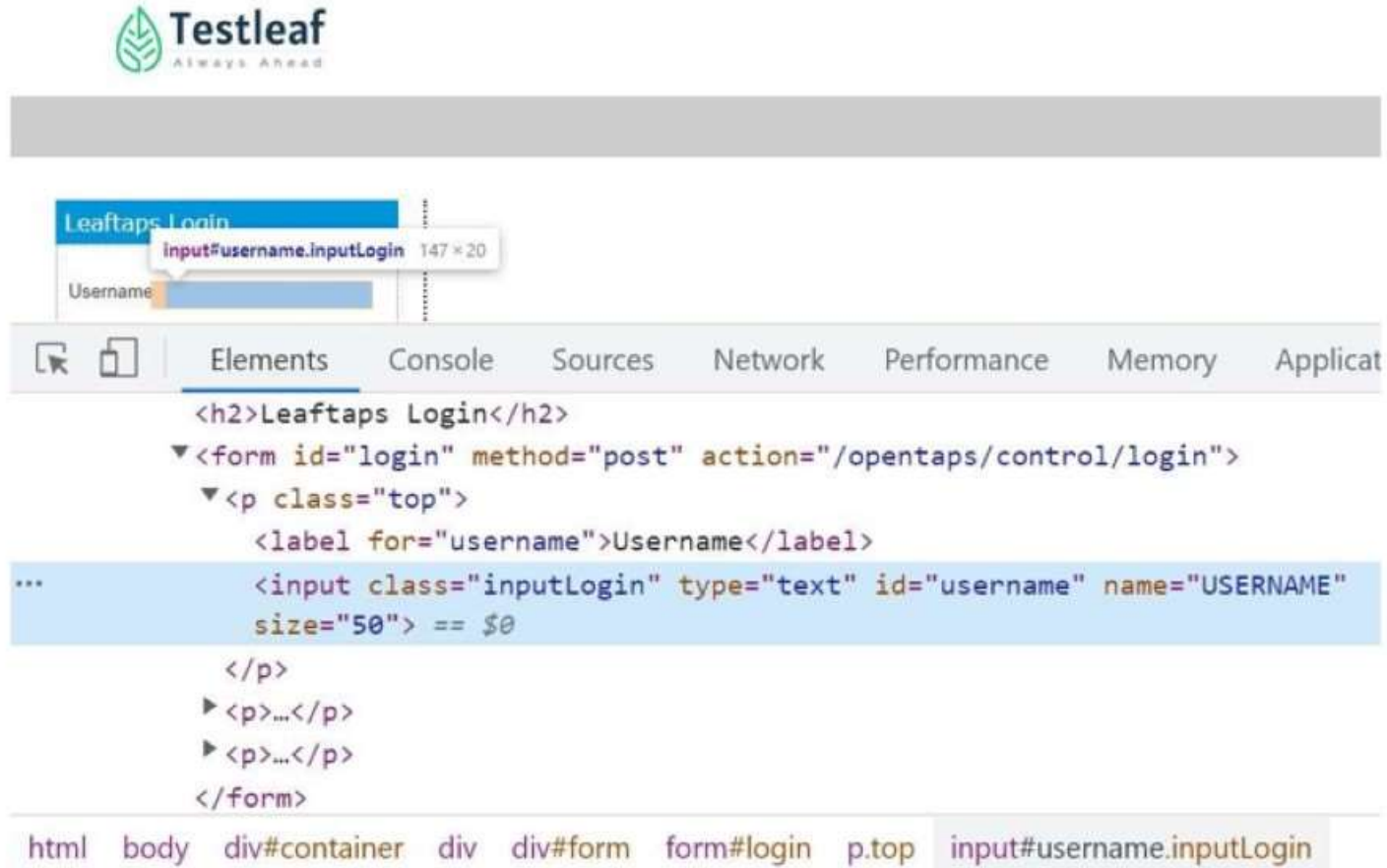
**When to use?**  
Not able to locate unique WebElement using Basic xpath.

How

How to write xpath?  
In relation with other elements

# Xpath (RelationShip based)

## LeafTaps- Login page



The screenshot shows the LeafTaps login page. At the top is the Testleaf logo with the tagline "Always Ahead". Below the logo is a grey header bar. The main content area contains the "LeafTaps Login" heading and a login form. The form has a "Username" label and a text input field. A tooltip above the input field shows the XPath expression "input#username.inputLogin" and its dimensions "147 x 20". Below the page is a browser developer tool with the "Elements" tab selected. The DOM tree shows the following structure:

```
<h2>LeafTaps Login</h2>
<form id="login" method="post" action="/opentaps/control/login">
  <p class="top">
    <label for="username">Username</label>
    ...
    <input class="inputLogin" type="text" id="username" name="USERNAME"
      size="50"> == $0
  </p>
  <p>...</p>
  <p>...</p>
</form>
```

The breadcrumb at the bottom of the developer tool shows the path: `html > body > div#container > div > div#form > form#login > p.top > input#username.inputLogin`.



# Xpath-Axes (type 1 of 8)

## *Parent To child* :

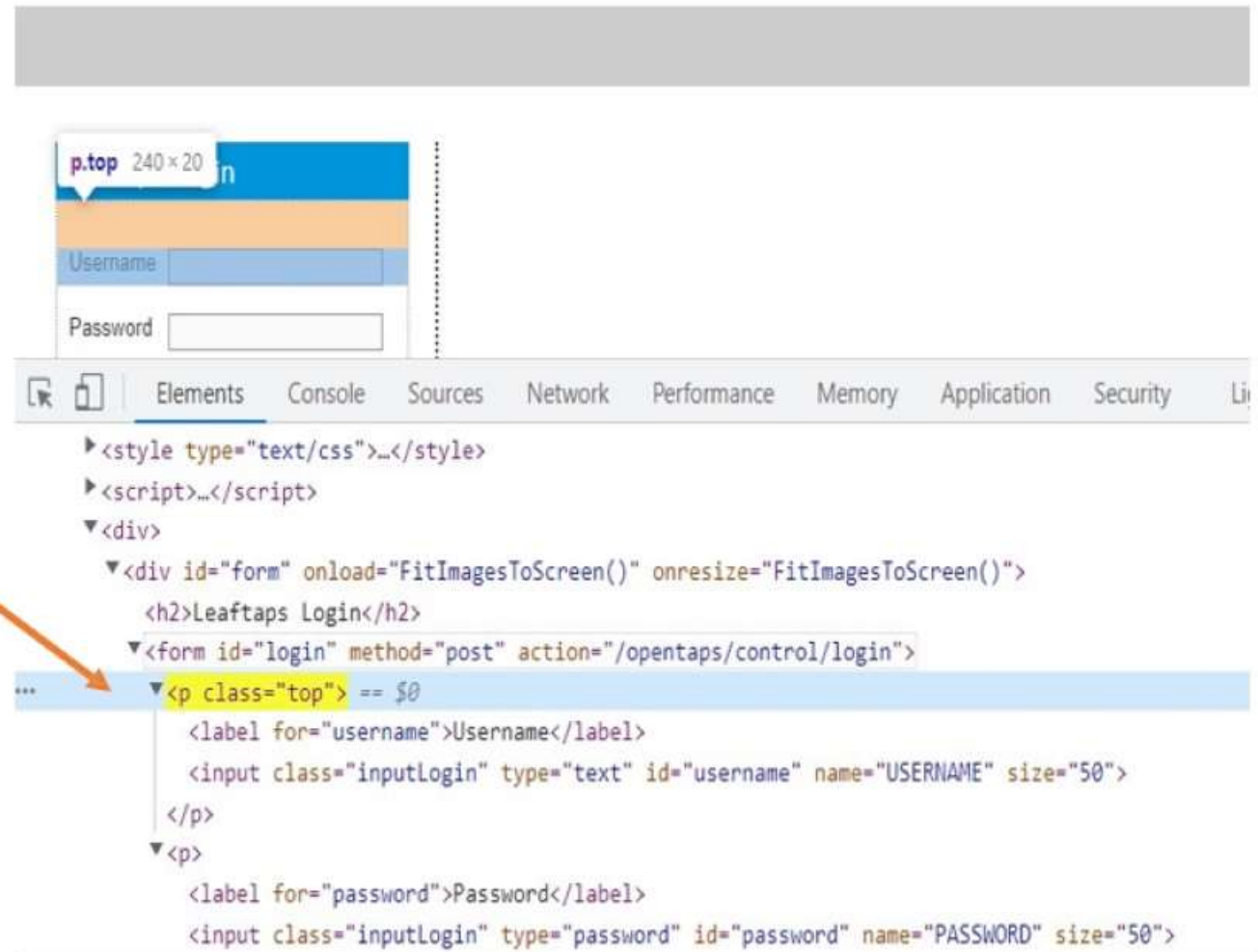
To reach the WebElement from `<form>` (parent tag) to `<p>` (child)

## Syntax:

`( parent relative xpath )/(tagName of child)`

## Example:

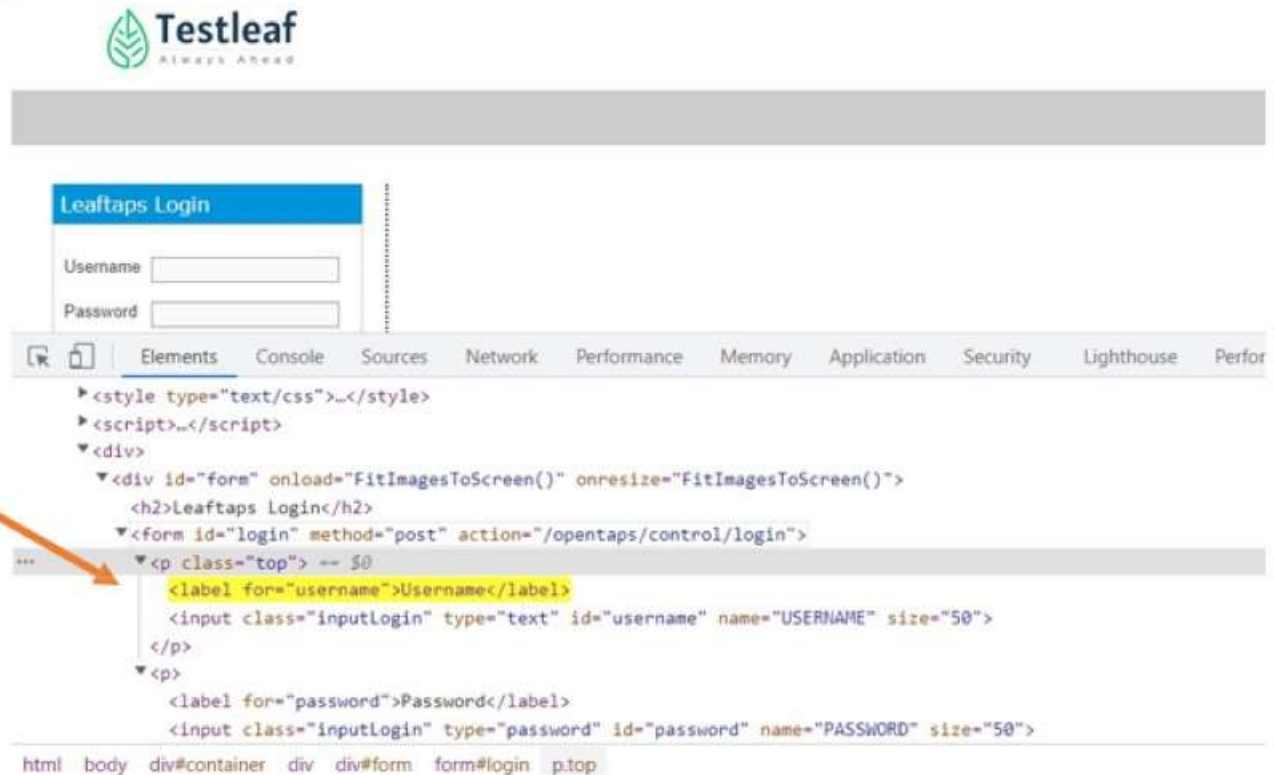
`//form[@id='login']/p`



# Xpath-Axes (type 2 of 8)

## **GrandParent to child Relation :**

To reach the WebElement from  
<form>(grandparent tag) to  
<label> (grand-child)



## **Syntax:**

**GrandParent relative xpath//tagName of grand child**

## **Example:**

**(//form[@id='login']//label)**

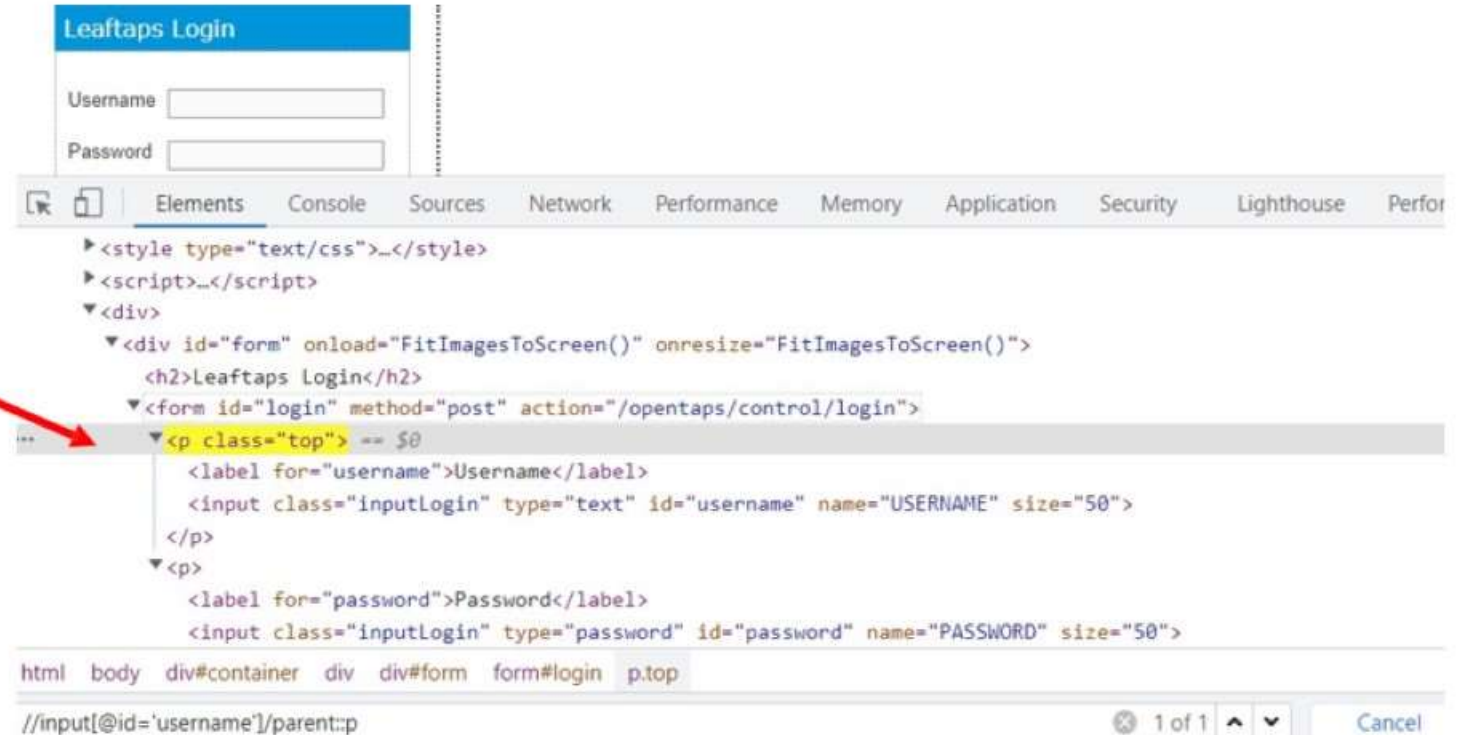
**Contd..**



# Xpath-Axes (type 3 of 8)

## *Child to parent :*

To reach the WebElement from **<p>(child)**  
to **<form> (parent)**



## Syntax:

Child relative xpath/parent::parent tagName

## Example

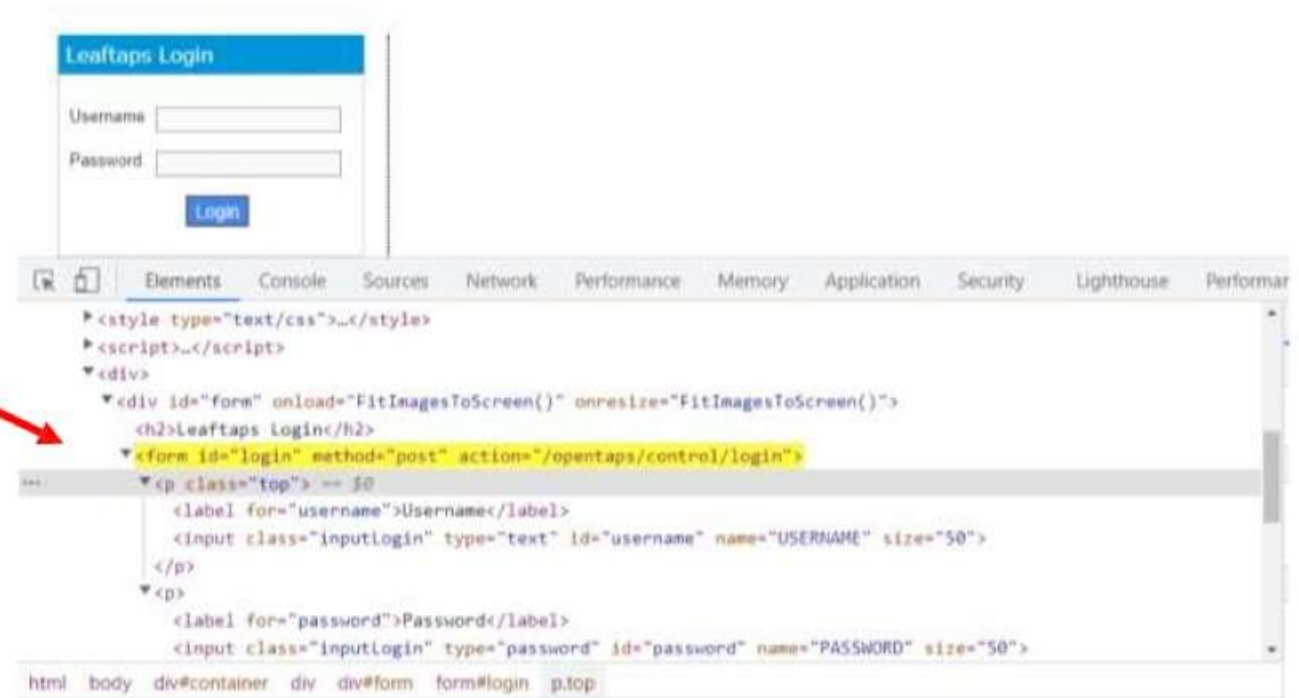
//input[@id='username']/parent::p

Contd..

# Xpath-Axes(type 4 of 8)

## *GrandChild to Grand parent :*

To reach the WebElement from **<label>(grand-child)** to **<form> (grand-parent)**, we have the keyword **ancestor**



## Syntax:

GrandChild relative `xpath/ancestor::tagName of grand parent`

## Example:

`//input[@id='password']/ancestor::form`

Contd..



# Xpath-Axes(type 5 of 8)

## *Elder Sibling to Younger Sibling :*

To reach the WebElement from <label> (child) to <label> (child) of same parent..

\_ we have the keyword  
**following-sibling** ( finding from top to bottom )



```
<style type="text/css">...</style>
<script>...</script>
<div>
  <div id="form" onload="FitImagesToScreen()" onresize="FitImagesToScreen()">
    <h2>Leaftaps Login</h2>
    <form id="login" method="post" action="/opentaps/control/login">
      <p class="top"> == $0
        <label for="username">Username</label>
        <input class="inputLogin" type="text" id="username" name="USERNAME" size="50">
      </p>
      <p>
        <label for="password">Password</label>
        <input class="inputLogin" type="password" id="password" name="PASSWORD" size="50">
      </p>
    </form>
  </div>
</div>
```

html > body > div#container > div > div#form > form#login > p.top

## Syntax:

(Elder sibling relative xpath)/following-sibling::tagName of younger sibling

## Example:

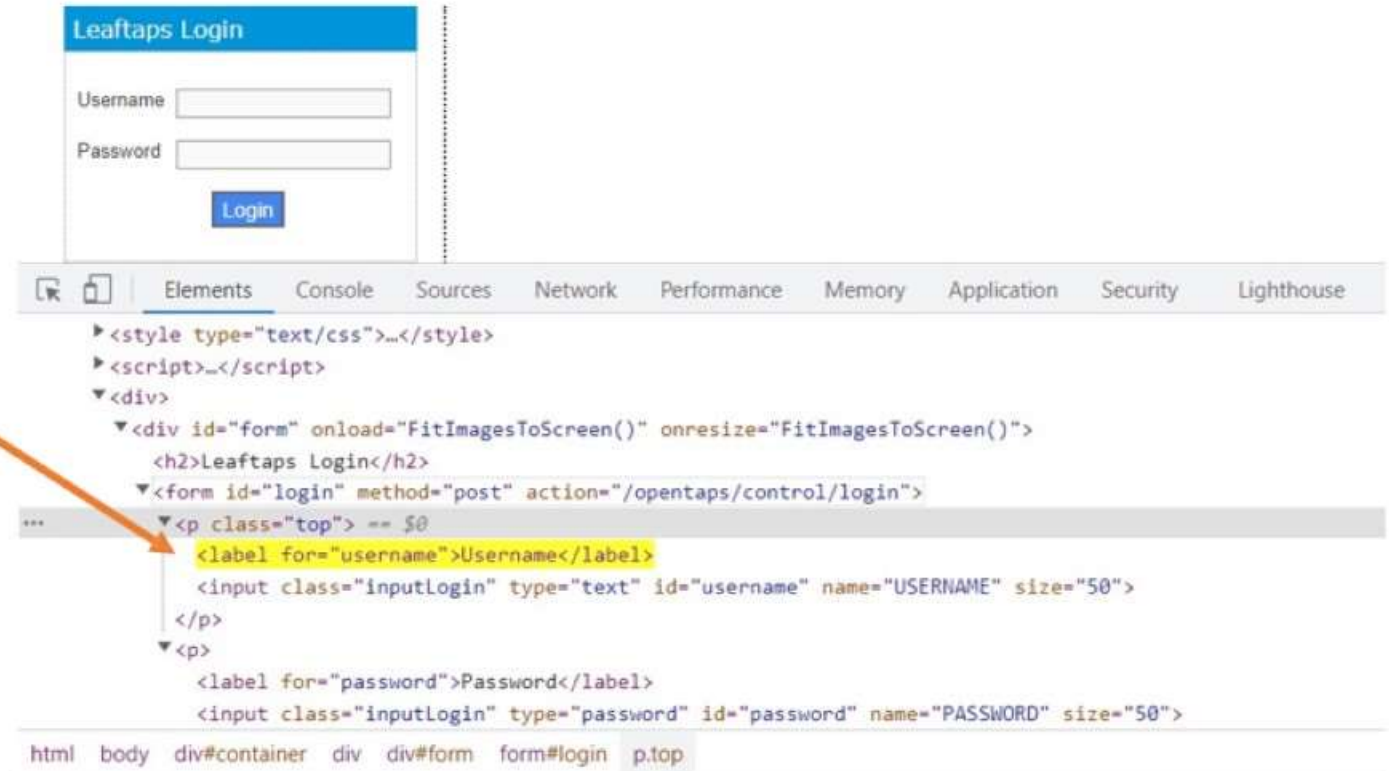
`//label[text()='Username']/following-sibling::input`

Contd..

# Xpath-Axes(type 6 of 8)

## Younger Sibling to elder Sibling:

To reach the WebElement from  
Second <label> (child) to first<label> (child)  
of same parent we have the keyword  
**preceding-sibling**( finding from bottom to top )



Syntax:

Younger sibling relative xpath/preceding-sibling::tagName of Elder sibling

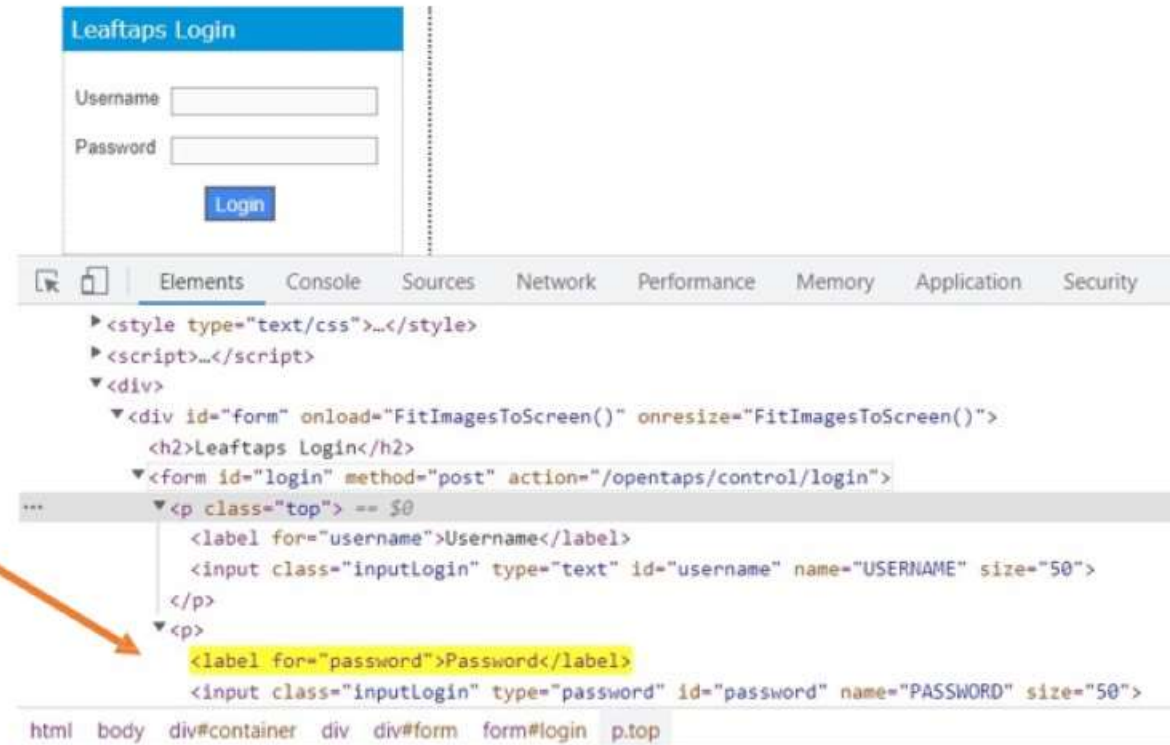
Example

`//input[@id='username']/preceding-sibling::label`

# Xpath-Axes(type 7 of 8)

## *Elder cousin to Younger cousin :*

To reach the WebElement from first parent child <label> (child) to second parent child <label> (child) we have the keyword **following** (finding from top to bottom)



Syntax:

Elder cousing relative xpath/following::tagName of younger Cousin

Example

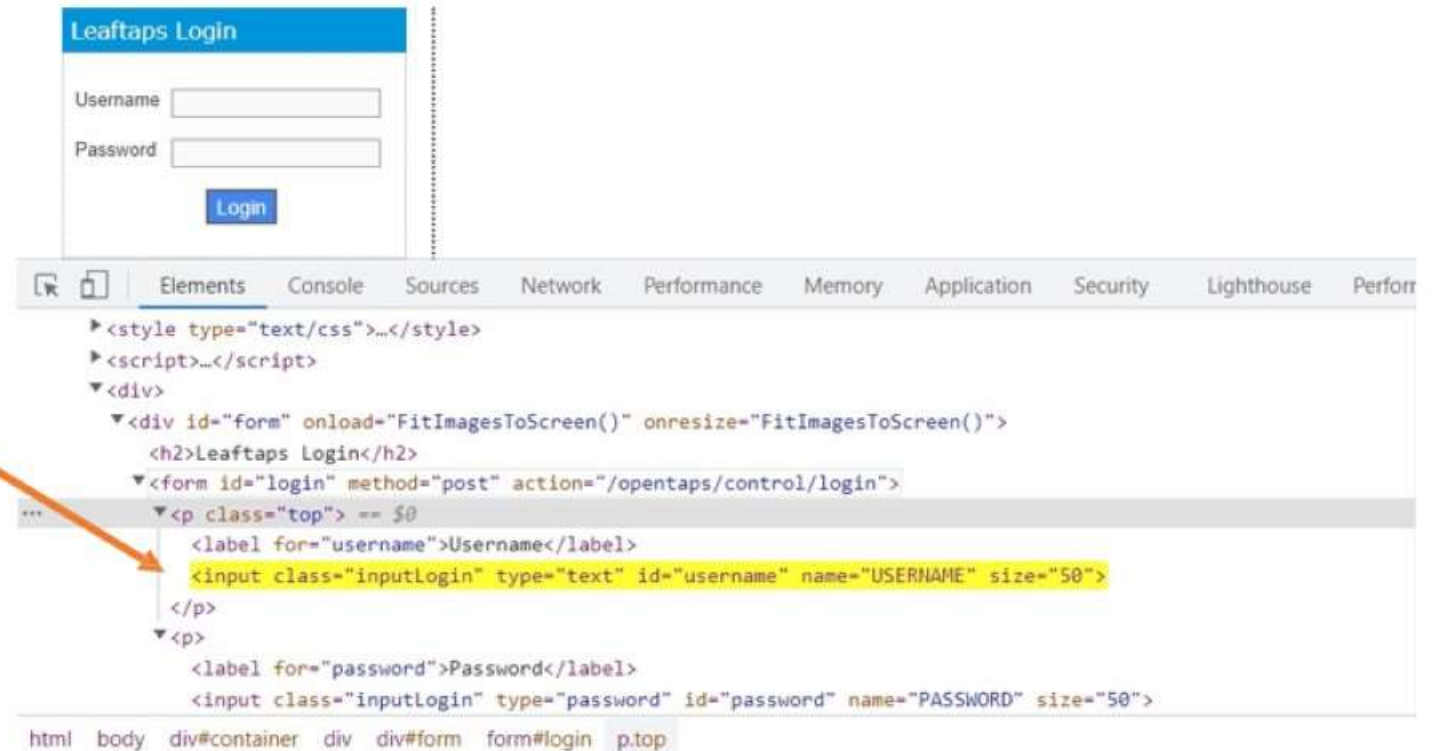
`//input[@id='username']/preceding-sibling::label`



# Xpath-Axes(type 8 of 8)

## *Younger cousin to Elder cousin :*

To reach the WebElement from second parent child <label> (child) to first parent child <label> (child) we have the keyword **preceding** ( finding from bottom to top )



Syntax:

Younger cousin `xpath/preceding::tagName of Elder cousin`

Example

`//label[text()='Password']/preceding::input`

# Quick Recall

- ✓ Axes Xpath-What , when, How?
- ✓ 8 types of Relationship based Xpath
  - Parent to Child & Viceversa
  - Grandparent to Grandchild & Viceversa
  - Sibling:
    - Elder to younger
    - Younger to Elder
  - Cousin
    - Elder to younger
    - Younger to Elder